



**INTRODUCTORY PHYSICS LABORATORY
DEPARTEMENT OF PHYSICS
FACULTY OF SCIENCE AND DATA ANALYTICS
INSTITUT TEKNOLOGI SEPULUH NOPEMBER
SURABAYA**



NAME :
STUDENT ID :
ASISTEN :
TANGGAL :

EXPERIMENT: MATHEMATICAL PENDULUM

Objective :

1. To determining the acceleration due to gravity on earth using a mathematical pendulum.

EXPERIMENT SUMMARY

INTRODUCTION

Oscillatory motion is one of the fundamental principles underpinning various natural and technological systems, such as spring vibrations, pendulum swings and waves. An object is said to be undergoing oscillatory motion if it moves back and forth periodically through its equilibrium point as a result of a restoring force that constantly directs the object back to its equilibrium position. One of the simplest forms of oscillatory motion is simple harmonic motion (SHM), which is motion in which the restoring force is proportional to the object's displacement from its equilibrium position. An understanding of simple harmonic motion forms the basis for studying various physical phenomena involving periodic motion [1].

One system that can be used to demonstrate simple harmonic motion is the mathematical pendulum. A mathematical pendulum is a system consisting of a point mass suspended from a light, inextensible string of a fixed length, in which air resistance and the mass of the string are neglected. When the pendulum is given a small displacement from its equilibrium position and then released, the force of gravity acts as a restoring force, causing the pendulum to swing periodically. However, the simple harmonic motion approximation only applies if the angle of displacement is small enough for the approximation $\sin \theta \approx \theta$ to hold [2].

The main quantities used to describe the motion of a mathematical pendulum are the period and the frequency. The period (T) is the time taken by the pendulum to complete one full swing, whilst the frequency (f) indicates the number of swings that occur per unit of time. These two quantities are related by the equation:

$$T = \frac{1}{f} \quad (1)$$

Consequently, the higher the frequency of the swing, the shorter the period, and vice versa. In a mathematical pendulum, the period is determined by the length of the string and the acceleration due to gravity on Earth; therefore, measuring the period can be used to determine the value of the acceleration due to gravity [1]. The torque acting on the pendulum arises from the force of gravity and thus satisfies the relationship:

$$\tau = -mgl \sin \theta \quad (2)$$

where (m) is the mass of the pendulum, (g) is the acceleration due to gravity on Earth, (l) is the length of the string, and (θ) is the angle of deflection. For small deflections, the approximation $\sin \theta \approx \theta$ applies, so the equation of motion becomes:

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0 \quad (3)$$

which is the differential equation governing simple harmonic motion. Solving this equation gives the period of the mathematical pendulum as:

$$T = 2\pi \sqrt{\frac{l}{g}} \quad (4)$$

and its frequency is given by:

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{l}} \quad (5)$$

this equation shows that the period of the swing depends only on the length of the string and the acceleration due to gravity on Earth, and is not affected by the mass of the pendulum. Furthermore, the period will increase if the length of the string is increased and will decrease if the acceleration due to gravity is greater. [3].

These principles are applied in the mathematical pendulum experiment to determine the value of the acceleration due to gravity on Earth by measuring the period of oscillation for various string lengths. The value of the acceleration due to gravity can be calculated directly using the period equation or determined by analysing the graph showing the relationship between the square of the period (T^2) and the string length (l). A comparison of these two methods provides an insight into the accuracy of the measurements and the effect of experimental uncertainty on the results obtained. The mathematical pendulum experiment not only demonstrates the characteristics of simple harmonic motion, but also provides an application of data analysis to determine one of the fundamental quantities in physics, namely the acceleration due to gravity on Earth [4].

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the mathematical pendulum

Apparatus	Quantity	Function
Mathematical pendulum set	1 set	Used as an oscillating system to observe simple harmonic motion.
Tape measure	1 piece	Used to measure the length of the pendulum string from the pivot point to the centre of mass of the pendulum.
Stopwatch	1 piece	Used to measure the time taken by the pendulum to complete one swing.

B. Procedure

The steps for this experiment are as follows :

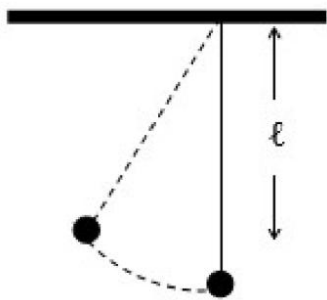


Figure 1. Illustration of a Mathematical Pendulum

1. Set up the apparatus as shown in Figure 1, with a string length of 100 cm.
2. Adjust the string length to the specified values: 20 cm, 35 cm and 50 cm.
3. Adjust the setup so that the tip of the pendulum is exactly in the centre (the string forms a vertical line). When measuring the length of the string, ensure that the bob extends into the measurement by half a finger’s width.
4. Displace the pendulum by a small amount (maximum 100) and release it. Ensure that the pendulum swings in a straight line, without rotating or hitting the table.
5. Record the time taken for five oscillations.
6. Repeat steps 1–5 five times as a series of repetitions.
7. Repeat steps 1–6 using different, pre-determined string lengths..

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement results obtained in the laboratory:

Table 2. Results of measurements using a mathematical pendulum and a large pendulum

Repitition	Period of 5 oscillations (s)		
	0,2 m	0,35 m	0,5m
1			
2			
3			
4			
5			
Average			

Table 3. Results of measurements using a mathematical pendulum and a small pendulum

Repitition	Period of 5 oscillations (s)		
	0,2 m	0,35 m	0,5m
1			
2			
3			
4			
5			
Average			

Approved by:

Laboratory Assistant

B. Analisis Data Hasil Perhitungan

(Calculate the acceleration due to gravity on earth)

The following is a sample calculation based on experimental data.

Sample Calculation:

Using the same method, the calculation above yields the following value for the acceleration due to gravity:

Table 4. Results of calculations using a mathematical pendulum and a large pendulum

Repetition	0,2 m		0,35 m		0,5 m	
	T(s)	g(m/s ²)	T(s)	g(m/s ²)	T(s)	g(m/s ²)
1						
2						
3						
4						
5						
Average						

Table 5. Results of mathematical pendulum calculations using a small pendulum

Repetition	0,2 m		0,35 m		0,5 m	
	T(s)	g(m/s ²)	T(s)	g(m/s ²)	T(s)	g(m/s ²)
1						
2						
3						
4						
5						
Average						

C. Graph

(Plot a graph of the length of the string (*l*) against the square of the period (*T*²), calculate the acceleration due to gravity from the graph, and explain the equation of the straight line)

D. Discussions

CONCLUSIONS

From this experiment, it can be concluded that:

- 1.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, Fundamentals of Physics Extended, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [2] D. C. Giancoli, Physics: Principles with Applications, 7th ed. Upper Saddle River, NJ, USA: Pearson Education, 2014.
- [3] R. A. Serway and J. W. Jewett, Physics for Scientists and Engineers with Modern Physics, 9th ed. Mason, OH, USA: Brooks/Cole, Cengage Learning, 2012.
- [4] M. Abdullah, Fisika Dasar I, Edisi Revisi. Bandung, Indonesia: Penerbit ITB, 2016.

PRELIMINARY ASSIGNMENT

1. Prove the equation for the period of a mathematical pendulum:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

by starting with the equation for the moment of force on a pendulum and using the small-angle approximation ($\sin \theta \approx \theta$) ?

2. Derive the equation expressing the acceleration due to gravity on Earth (g) as a function of the length of the string (l) and the period of oscillation (T), based on the equation in question 1!
3. In the equation for the period of a mathematical pendulum, the mass of the bob is not included. Explain why mass does not affect the period of oscillation?
4. Explain why ($\sin \theta \approx \theta$) required in deriving the period equation for a mathematical pendulum. How does this affect the results of the calculations if the initial displacement is sufficiently large?
5. Derive the frequency equation for a mathematical pendulum from the period equation, then explain the relationship between the frequency (f) and the length of the pendulum string (l). How does the frequency change if the length of the string is increased?